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Bachelor's Thesis

# Measuring Drift in Therapeutic AI: A Stability-Based Evaluation of Sovereign LLMs in Nightmare Therapy

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# Declaration of Authorship

The content of this thesis represents my own knowledge, my own understanding and my own perspective on the topic. In case artificial intelligence tools were used, their way and purpose of usage has been made transparent. Moreover, I have cited all my sources in accordance with academic standards. I am ready and able to explain and defend the positions developed in this thesis. This thesis has not been submitted, either in part or whole, at this or any other university.

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# Abstract

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# 1. Introduction

## 1.1. Nightmare Disorder and Imagery Rehearsal Therapy

Nightmare disorder affects an estimated 2–5% of the general adult population (Li et al., 2010; Sandman et al., 2013; Schredl, 2010) and frequently co-occurs with PTSD and depression (Morgenthaler et al., 2018). The recommended treatment, Imagery Rehearsal Therapy (IRT), is a structured five-stage protocol in which the patient rewrites a recurring nightmare under therapist guidance (Krakow & Zadra, 2006). A persistent shortage of trained therapists limits the availability of IRT (Wittchen et al., 2011), making scalable, AI-assisted delivery a practical goal. However, the rewriting stage requires active strategy decisions from the therapist, and an AI agent that selects different strategies on repeated runs of the same case is clinically unreliable regardless of how capable any single response may be.

## 1.2. AI-Assisted Psychotherapy

Large language models are increasingly deployed in mental health, from empathy-based conversational tools to protocol-driven agents that deliver structured interventions such as cognitive behavioural therapy (Stade et al., 2024; Q. Zhang et al., 2025). reDreamAI is the deployment context for this study: a consumer-facing IRT chatbot that guides users through all five protocol stages. It is not a research prototype but a real deployment target where reliability is non-negotiable.

This deployment context creates a specific problem. A general-purpose assistant has no therapeutic obligation; response variability is acceptable and even desirable. A protocol-driven agent, by contrast, must make consistent strategic decisions. If the model selects different rescripting strategies for the same patient on different runs, the patient receives inconsistent care. Unlike rule-based chatbots whose responses can be reviewed once before deployment (Fitzpatrick et al., 2017), an LLM agent generates a new response on every run. Each run is a new draw from the output distribution, so stability must be evaluated repeatedly rather than verified once.

IRT targets nightmares rather than psychiatric diagnoses, which lowers the clinical risk relative to full psychotherapy. Ethical considerations are discussed in section 6.5 and section 7.2. The EU AI Act classifies autonomous therapeutic AI as high-risk and requires auditability (Aboy et al., 2024; van Kolfschooten & van Oirschot, 2024). Data sovereignty requirements push toward EU-resident models (Dale, 2025), introducing a constraint on model selection discussed in section 3.8.

## 1.3. The Evaluation Gap

Standard NLP benchmarks measure what a model *can* do, not whether it does the same thing reliably. A model can score well on a single evaluation while producing different therapeutic strategies on repeated runs of the same input. Capability and stability are distinct properties, but current benchmarks conflate them.

The closest existing work is BOLT (Chiu et al., 2024), which evaluates LLM therapist behaviour using 13 behavioural codes from motivational interviewing. BOLT measures whether a model applies appropriate therapeutic techniques (quality). This study measures a different property: whether a model applies the *same* techniques across repeated runs (stability). BOLT evaluates single responses; this study evaluates the distribution of responses across twenty independent trials per condition.

Other therapeutic AI evaluations are qualitative, post-hoc, or single-session (Talat et al., 2024). They cannot detect cross-run inconsistency because they do not present the same input multiple times. Systematic, reproducible in silico trials offer a way to measure stability at scale without requiring human participants.

## 1.4. Research Objectives

This thesis addresses the following research question:

> How consistent are LLM therapeutic responses across independent runs when given the same patient context within a structured IRT protocol?

Two contributions follow from this question.

**Contribution 1.** A three-level evaluation framework that measures stability at three layers: *plan consistency* (do strategy selections repeat?), *response similarity* (do the generated texts convey the same meaning?), and *plan-response alignment*

(does the model do what it declared?). Each layer captures a different aspect of therapeutic stability, and their combination reveals divergence patterns that no single metric can detect.

**Contribution 2.** A cross-model comparison that applies this framework to sovereign, proprietary, and open-weight models under controlled temperature conditions.

The study isolates the IRT rescripting phase because it is the stage that requires active strategy decisions. Isolating a single stage eliminates stage-routing confounds and allows any measured instability to be attributed to the therapist model rather than to upstream variation in the conversation. chapter 3 describes the experimental design in detail.

# 2. Background

## 2.1. Imagery Rehearsal Therapy

Nightmare disorder affects an estimated 2–5% of the adult population (Li et al., 2010; Sandman et al., 2013; Schredl, 2010) and shows strong comorbidity with Post-Traumatic Stress Disorder (PTSD) and depression (Morgenthaler et al., 2018). For patients with recurrent nightmares, the distress extends beyond the dream itself: sleep avoidance, daytime fatigue, and heightened anxiety create a cycle that standard sleep hygiene cannot break. IRT is the recommended treatment for this condition (Krakow & Zadra, 2006; Morgenthaler et al., 2018).

IRT follows a structured five-stage protocol. In the *recording* stage, the patient describes the nightmare in detail. In the *rewriting* stage, the patient and therapist collaboratively transform the nightmare narrative to reduce distress and increase a sense of mastery. The remaining stages (*summary*, *rehearsal*, and *final*) consolidate the new narrative and prepare the patient to rehearse it before sleep. Krakow and Zadra (2006) describe the rewriting instruction as deliberately open-ended: patients are told to "change the nightmare any way you wish."

The rewriting stage is the cognitive core of the protocol. Unlike recording (which requires listening) or rehearsal (which requires repetition), rewriting demands active therapeutic strategy decisions. The therapist must guide the patient toward a specific rescripting mechanism: confronting the threat, introducing protective elements, reframing the narrative, or transforming the dreamer's role within it. These choices are not arbitrary. Clinical research has identified distinct rescripting strategies with different prevalence rates in successful treatment. Harb et al. (2012) coded five categories from combat veteran rescripts: alternative endings (58%), positive image insertion (23%), threat transformation (13%), reality markers (10%), and distancing (8%). Germain et al. (2004) proposed a complementary framework through the Multidimensional Mastery Scale, which classifies rescripting outcomes along physical, social, environmental, emotional, mystical, and avoidance dimensions.

Treatment fidelity instruments provide the clinical standard for measuring whether a therapist delivers an intervention correctly. The IRT literature draws on two established frameworks: the ENACT rating scale (Kohrt et al., 2015) and the NIH BCC recommendations (Bellg et al., 2004). Both grade therapist adherence on a ternary scale: *absent*, *partial*, or *implemented*. This gradation captures clinically meaningful differences. A therapist who briefly mentions a rescripting strategy without developing it (partial) delivers a different intervention from one who fully guides the patient through it (implemented).

This ternary framework provides a scoring foundation that transfers directly to computational evaluation (see subsection 3.7.3). However, existing fidelity instruments assume a human clinician. They have not been adapted for Large Language Model (LLM)-based agents, where the therapist is a probabilistic text generator rather than a trained professional.

## 2.2. Treatment Fidelity in LLM-Based Interventions

The deployment of LLMs in mental health has expanded from empathy tools and symptom screeners to protocol-driven therapeutic agents. Stade et al. (2024) survey this trajectory and argue that LLMs could fundamentally reshape behavioural healthcare delivery, from diagnostic support to full psychotherapeutic interventions. Early systems such as Woebot (Fitzpatrick et al., 2017) delivered Cognitive Behavioral Therapy (CBT) through rule-based dialogue trees where every response was pre-authored and reviewed before deployment. For these systems, treatment fidelity is a design property: the chatbot cannot deviate from its script.

LLM-based agents operate differently. Each response is a fresh sample from the model's output distribution. A single model can produce high-quality, protocol-adherent responses on one run and strategically inconsistent ones on the next. This distinction is central to the treatment fidelity literature. Mowbray et al. (2003) formalise fidelity into two components: *adherence* (doing the prescribed things) and *competence* (doing them well). For rule-based chatbots, adherence is built in by construction. For LLM-based protocol agents, adherence becomes probabilistic and must be measured across runs rather than assumed from a single review.

LLM-based agents also face general failure modes such as hallucination, stage confusion, and persona inconsistency. These are well-documented concerns for clinical AI but fall outside the scope of this study. The present work focuses on a narrower problem: the probabilistic nature of LLM outputs creates drift even in

identical therapeutic contexts. When the same patient presents the same nightmare, stochastic sampling can produce different strategy selections, different therapeutic responses, and mismatches between declared and executed strategies. Each of these failure modes corresponds to a distinct layer of the generation process and requires a targeted evaluation method. Variable strategy selection maps to plan-level measurement (subsection 3.7.1), variable response content to semantic comparison (subsection 3.7.2), and plan-response mismatch to alignment verification (subsection 3.7.3).

Regulatory frameworks reinforce the need for stability measurement. The European Union (EU) AI Act classifies autonomous therapeutic AI as high-risk, imposing requirements for auditability, documentation, and risk management (Aboy et al., 2024; van Kolfschooten & van Oirschot, 2024). The Medical Device Regulation adds further constraints on software used in clinical contexts. For EU-based deployment scenarios, data sovereignty requirements push toward self-hosted, EU-resident models (Dale, 2025), constraining the model selection space (see section 3.8).

Despite these requirements, stability remains largely unmeasured. Prior therapeutic AI evaluation is qualitative, post-hoc, or single-session. Chiu et al. (2024) represent the closest prior work with their BOLT framework, which assesses LLM therapist behaviour across 13 behavioural codes drawn from motivational interviewing. BOLT asks whether the model does the right thing. The present study asks a different question: whether the model does the *same* thing across repeated runs. Existing frameworks, including BOLT, evaluate single responses rather than the distribution of responses across repeated trials.

## 2.3. Stochastic Behaviour in Language Models

At temperatures above zero, LLM token selection is probabilistic. The softmax distribution over the vocabulary is divided by the temperature parameter before sampling, so higher temperatures flatten the distribution and lower temperatures sharpen it toward the most likely token. At $T = 0$, the model selects the highest-probability token deterministically (greedy decoding). At $T > 0$, tokens are sampled from the resulting distribution, introducing variance into the output.

This per-token variance compounds through autoregressive generation. Each token is conditioned on all preceding tokens, so a single divergent token early in a response can cascade into a completely different therapeutic strategy. A model that selects “confront” rather than “imagine” at one token position may generate an en-

tirely different rescripting approach in the subsequent response. This *autoregressive cascading* means that the clinical impact of stochastic sampling is not proportional to the per-token entropy but can amplify through the response.

The clinical consequence is concrete: the same patient presenting the same nightmare on two different days may receive strategically inconsistent care. In a general-purpose assistant, this variability is benign or even desirable. In a manualized therapeutic protocol, where the therapist is expected to deliver consistent, evidence-based interventions, it represents a potential clinical failure.

Prior work on LLM stability addresses different questions than therapeutic consistency. Factual consistency studies measure whether models produce the same answers to knowledge questions across runs. Output diversity research quantifies lexical and semantic variation in creative generation. Benchmark reproducibility work examines whether evaluation scores are stable across random seeds. These lines of research share the concern for cross-run variation but do not address whether a model makes the same *clinical decisions* when presented with identical patient contexts.

The mechanical source of instability is well understood. What remains unquantified is its impact on structured therapeutic decision-making. How much does temperature-driven variance cost in clinical consistency? This question motivates the temperature sweep in the experimental design (section 3.9), which tests five points from $T = 0.0$ (deterministic) to $T = 0.6$ (typical vendor default for open-weight models).

## 2.4. Evaluation Metrics for Text Generation

Measuring consistency between text outputs requires a similarity metric. The choice of metric determines what kind of consistency is captured, and standard options fail in different ways for therapeutic language.

Surface-level metrics such as BLEU and ROUGE measure token overlap between texts. In therapeutic contexts, these metrics produce misleading results in both directions. Two responses that use identical words to describe different strategies score high on surface overlap despite being clinically divergent. Conversely, two responses that implement the same strategy through different phrasing score low despite being therapeutically equivalent. Therapeutic paraphrasing is the norm rather than the exception: a competent therapist will naturally vary their language

while maintaining the same strategic intent. Surface metrics penalise this valid variation.

BERTScore (T. Zhang et al., 2020) addresses this limitation by computing similarity in contextual embedding space rather than token space. Each token in the candidate text is matched to the most similar token in the reference text using cosine similarity over contextualised embeddings, producing precision, recall, and F1 scores. The metric captures semantic equivalence independent of surface form: two sentences that mean the same thing but use different words receive high scores.

The choice of embedding model affects BERTScore's behaviour. Among more than 130 available models, those fine-tuned on Natural Language Inference (NLI) tasks achieve the highest correlation with human judgements of semantic equivalence. DeBERTa-XLarge-MNLI (He et al., 2021) achieves the highest WMT16 human correlation ($r = 0.778$) and is selected for Method 2 in this study (subsection 3.7.2). Its disentangled attention mechanism handles the surface-form variation typical of therapeutic language, where valid paraphrasing is expected.

BERTScore captures semantic equivalence but cannot assess strategic intent. A metric that reports high similarity between two responses does not reveal whether both responses implement the same therapeutic strategy. The LLM-as-judge paradigm addresses this gap. By instructing a separate LLM to evaluate whether a response implements a declared strategy, it becomes possible to assess criteria that no scalar metric can capture: "does this response implement a confrontation strategy?" requires understanding both the strategy definition and the response content.

LLM-as-judge approaches carry known biases. Self-preference bias causes models to rate outputs from their own family more favourably. Position bias leads to higher scores for responses presented first. Verbosity bias rewards longer responses regardless of quality (Zheng et al., 2023). These biases can be mitigated through cross-model judging (the judge evaluates a different model family), structured rubrics, and explicit justification requirements. The mitigation strategy adopted in this study is detailed in subsection 3.7.3.

No single metric covers what clinical stability requires. Embedding similarity captures semantic equivalence but misses strategic intent. LLM judges can assess intent but introduce their own reliability concerns. Surface metrics capture neither. Existing therapeutic AI evaluations rely on a single metric type, leaving the gap between semantic equivalence and strategic intent unaddressed. This motivates the three-method design presented in chapter 3.

# 3. Methods

## 3.1. System Overview and Design Rationale

The evaluation system separates dialogue creation from stability measurement through two decoupled stacks. The *generation stack* produces synthetic therapy sessions by running a multi-agent dialogue loop. The *evaluation stack* measures the stability of therapeutic responses by presenting frozen conversation histories to the model under test and collecting multiple independent outputs.

This separation is a deliberate design choice. If the evaluation stack generated its own dialogue context, any measured instability could reflect upstream randomness in the patient or routing agents rather than genuine variation in the therapist model. By freezing the conversation history before evaluation begins, the system ensures that every trial receives identical input. Any observed inconsistency is attributable solely to the model under test.

The end-to-end data flow proceeds in five stages: patient vignettes feed the generation stack, which produces complete therapy sessions. These sessions are sliced at rewriting-turn boundaries to create frozen histories. The evaluation stack presents each frozen history to the therapist model across multiple independent trials, collecting both a strategy declaration and a therapeutic response per trial. Finally, an aggregation pipeline aligns all metrics across conditions for comparative analysis.

## 3.2. Dialogue Generation

The generation stack orchestrates a three-agent loop that produces naturalistic IRT sessions. Each turn proceeds through three agents in sequence:

1. The **patient agent** simulates a clinical presentation based on a Brief Dreamer Interview (BDI)-profiled vignette. Each vignette defines the patient's nightmare content, emotional presentation style, and engagement pattern.

2. The **router agent** classifies the current IRT stage from the conversation history, determining whether the session is in the recording, rewriting, summary, rehearsal, or final phase.

3. The **therapist agent** generates a stage-appropriate therapeutic response, guided by the router's classification and a stage-specific system prompt.

This three-agent architecture separates clinical reasoning from stage classification and patient simulation, matching the deployment architecture of reDreamAI. The router's stage classification controls which system prompt the therapist receives, ensuring that responses are appropriate to the current protocol phase.

The vignette design is informed by anonymised session data from a prior reDreamAI efficacy study. Because data protection requirements prevent using real patient transcripts directly, synthetic patient profiles were created and validated against published clinical research on simulated patients.

Six vignettes cover the range of clinical presentation difficulty expected in IRT deployment: *anxious*, *avoidant*, *cooperative*, *resistant*, *skeptic*, and *trauma*. These profiles range from a cooperative patient who engages readily with rescripting instructions to a trauma patient whose nightmare content triggers intense emotional responses during the session. The design draws on prior synthetic patient work. Z. Wang et al. (2024) demonstrate that expert-defined behavioural principles produce clinically plausible simulated patients. R. Wang et al. (2024) provide cognitive conceptualisation diagrams and conversational styles (direct, resistant, expressive, minimal, deviating, agreeable) that parallel the six archetypes used here. Six profiles sample the difficulty range but do not claim exhaustive coverage of patient variability (see section 7.2).

The generation stack traverses the full five-stage IRT protocol, producing complete therapy sessions that are then sliced to create evaluation entry points.

## 3.3. Frozen History Design

Frozen conversation histories serve as deterministic evaluation entry points. Each vignette's complete therapy session is sliced at rewriting-turn boundaries, where each slice ends after a complete therapist-patient exchange. Every trial, across all models and temperatures, receives the identical frozen history as input. This design eliminates context variation as a confound: if two trials produce different therapeutic

responses, the difference is attributable to the model's stochastic sampling, not to differences in the input conversation.

A preliminary analysis tested Mistral Large 3 at five slice depths across all six vignettes at $T = 0.075$ and $T = 0.15$. Jaccard consistency was lowest at the second slice (mean $J = 0.77$ at $T = 0.075$) and rose to near-ceiling values from the third slice onward ($J > 0.93$). The main experiment evaluates at the second rewriting-turn boundary (`slice_2`), making the measured stability a conservative estimate. Slice depth effects are reported in section 5.6.

## 3.4. Evaluation Stack

The evaluation stack bypasses the router and injects the rescripting prompt directly into the therapist model. Because the IRT stage is known a priori (all frozen histories are sliced within the rewriting phase), routing is unnecessary and would introduce an additional source of variation. Bypassing it isolates the therapist model from stage-classification error.

Each evaluation trial uses *fused generation*: the model produces a `<plan>` block declaring one to two therapeutic strategies, followed by the therapeutic response, in a single call. The plan tokens condition the response tokens in a Chain-of-Thought (CoT)-style generation, so the response is directly conditioned on the declared strategy. This fused approach enables the alignment analysis in Method 3 (subsection 3.7.3): because plan and response are generated in a single pass, any mismatch between declared intent and executed strategy reflects the model's own inconsistency rather than a decoupling artefact.

Twenty independent trials are collected per condition, yielding $\binom{20}{2} = 190$ pairwise comparisons. This trial count balances statistical power with compute cost: 190 pairs provide robust coverage of the pairwise similarity distribution while keeping the total experiment within budget.

## 3.5. The Plan Mechanism

The `<plan>` block raises a methodological question. CoT prompting (Wei et al., 2022) improves output structure by asking models to show their reasoning before answering. However, the faithfulness literature demonstrates that CoT explanations often do not reflect the model's actual computation. Turpin et al. (2023) show that biasing cues in the prompt alter model behaviour without appearing in the

CoT trace. Lanham et al. (2023) find that early CoT tokens can be truncated or replaced without affecting the final answer, suggesting that the reasoning trace is reconstructed post-hoc rather than used for computation.

If a model declares its therapeutic strategy before responding, that declaration may be a post-hoc rationalisation rather than a window into its decision-making process. This study resolves the tension by framing the `<plan>` block as *declared intent* rather than a reasoning trace. The plan captures what the model commits to, not what it internally computes. Whether or not the declaration reflects genuine reasoning, a practical consequence holds: if strategy declarations vary across runs, clinical responses will also vary. Variable declared intent predicts variable therapeutic behaviour, which is the clinically relevant property.

This framing sidesteps the faithfulness debate entirely while preserving three concrete uses. First, the declared strategies enable strategy-level consistency scoring (Method 1, subsection 3.7.1). Second, comparison between declared strategies and the executed response enables alignment verification (Method 3, subsection 3.7.3). Third, the plan provides a human-readable summary of the model's intended approach, supporting clinical oversight in deployment.

## 3.6. Strategy Taxonomy Development

A fixed taxonomy constrains the `<plan>` block to a predefined set of therapeutic strategies. This constraint is necessary for quantitative evaluation: free-text plans cannot be aggregated into set-similarity scores. The taxonomy evolved through four iterations to resolve a severe distribution skew that rendered early versions unusable.

The initial taxonomy (v1) defined eight categories including `empowerment` and `mastery`. Across 1,272 strategy picks, these two categories consumed 87.7% of all selections, while three categories (`social_support`, `sensory_modulation`, `gradual_exposure`) were never chosen. The root cause was a goal-mechanism conflation: both `empowerment` and `mastery` describe a therapeutic *goal* (the patient gains control) rather than a specific *mechanism* of change. Since every nightmare rescripting intervention serves the goal of increased mastery, the model defaulted to these labels regardless of the actual strategy employed.

Merging the two categories into a single `agency` category (v2) amplified the problem: the merged category achieved 100% dominance. A single super-category that applies to every rescripting scenario provides no discriminative signal.

Table 3.1.: Strategy taxonomy (v4). Each category describes a rescripting mechanism rather than a therapeutic goal.

| Category | Mechanism |
|---|---|
| `confrontation` | The dreamer directly faces, fights, or stands up to the threat |
| `self_empowerment` | The dreamer gains a new ability or inner strength |
| `safety` | A physical barrier, shelter, or protective figure shields the dreamer |
| `cognitive_reframe` | Existing dream elements change meaning |
| `social_support` | A new person, companion, or ally is introduced |
| `sensory_modulation` | Sensory qualities shift, including internal somatic calming |

The resolution in v3 was to split `agency` along the mechanism dimension. `Confrontation` describes external action toward the threat (the dreamer fights, challenges, or resists). `Self_empowerment` describes internal transformation within the dreamer (gaining abilities, growing stronger). These two categories are orthogonal: a dreamer who confronts a monster uses a different mechanism from one who discovers inner strength, even though both gain mastery in the outcome.

A final revision (v4) merged `emotional_regulation` into `sensory_modulation` after smoke testing showed that `emotional_regulation` appeared in only 1 of 120 trials (0.4%). In single-turn rescripting, internal calming manifests as sensory change (the dreamer feels calm, the environment becomes peaceful), making the boundary between the two categories clinically indistinct at this granularity.

The final six-category taxonomy captures concrete therapeutic mechanisms:

These categories are grounded in two clinical frameworks. Germain et al. (2004) identify six mastery dimensions in successful nightmare rescripting (physical, social, environmental, emotional, mystical, avoidance), all of which map to the taxonomy except avoidance. Harb et al. (2012) provide empirical rescripting frequencies from combat veterans, with all therapeutically productive categories covered. Avoidance and distancing are deliberately excluded: Harb et al. (2012) found that violence in rescripted dreams predicted worse treatment outcomes, and Krakow and Zadra (2006) frame IRT as "changing the nightmare" rather than escaping it. Reality markers (lucid dreaming cues) fall outside the scope of single-turn rescripting.

The iterative development process is documented here because taxonomy design directly shapes measured consistency. A coarse taxonomy inflates agreement scores (every trial picks the same super-category). A fine-grained taxonomy deflates them

(minor strategy differences register as disagreements). The sensitivity of consistency metrics to category design is revisited in the discussion (section 6.5).

## 3.7. Evaluation Metrics

Three evaluation methods target distinct layers of the generation process. No single metric captures the full picture of clinical stability (section 2.4). Method 1 asks whether the model makes consistent therapeutic decisions. Method 2 asks whether it produces consistent therapeutic responses. Method 3 asks whether it does what it says it will do.

### 3.7.1. Method 1: Plan Consistency

Method 1 measures whether the model selects the same therapeutic strategies across independent trials. Strategy sets are extracted from the `<plan>` block of each trial output using the fixed taxonomy from Table 3.1. Pairwise Jaccard similarity quantifies agreement between strategy sets:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{3.1}$$

The mean Jaccard score is computed over all $\binom{20}{2} = 190$ trial pairs per condition. Jaccard is appropriate because strategy combinations are unordered sets: a model that declares `confrontation / cognitive_reframe` has made the same clinical decision as one that declares `cognitive_reframe / confrontation`.

Because Jaccard operates on discrete sets of one to two items drawn from six categories, it takes only a small number of distinct values: $\{0, 0.33, 0.5, 0.67, 1.0\}$. This discreteness means Jaccard scores are ordinal rather than continuous, and descriptive statistics report medians and interquartile ranges rather than means and standard deviations.

Plan validity rate serves as an upstream quality gate. A malformed `<plan>` block that cannot be parsed into valid strategy identifiers is excluded from the Jaccard computation. The validity rate reports the proportion of trials with parseable plans, indicating whether the model can follow the structured output format at all.

### 3.7.2. Method 2: Response Similarity

Method 2 measures whether the model produces therapeutically equivalent responses across trials, independent of strategy declarations. Pairwise BERTScore F1 is com-

puted over response texts (with plan blocks excluded) across the same 190 trial pairs.

The embedding model is DeBERTa-XLarge-MNLI (He et al., 2021). This model achieves the highest human correlation ($r = 0.778$) among more than 130 candidates on WMT16 segment-level evaluation. Its NLI fine-tuning aligns with the semantic equivalence question: determining whether two therapeutic responses convey the same clinical content, regardless of surface phrasing. Its disentangled attention mechanism handles the surface-form variation typical of therapeutic language, where a competent response naturally varies in wording while maintaining the same clinical intent.

Surface metrics (BLEU, ROUGE) are not used. As established in section 2.4, token-overlap metrics penalise valid therapeutic paraphrasing and cannot distinguish semantically equivalent from semantically divergent responses when surface forms differ.

### 3.7.3. Method 3: Plan-Response Alignment

Method 3 measures whether the model's therapeutic response implements its declared strategy. A separate LLM judge scores each declared strategy on a ternary scale:

- **0 (absent):** The strategy is not reflected in the response.
- **1 (partial):** The strategy is mentioned or briefly addressed but not fully developed.
- **2 (implemented):** The strategy is clearly and substantively present in the response.

This ternary scale is borrowed from the clinical fidelity literature (section 2.1). The ENACT (Kohrt et al., 2015) and NIH BCC (Bellg et al., 2004) frameworks use the same three-level gradation (absent, partial, implemented) to assess whether a human therapist delivers an intervention correctly. Applying the same scale to LLM outputs maintains continuity with established clinical evaluation practice.

The judge model is Gemini 3.1 Pro, accessed through Google AI Studio at $T = 1.0$. Gemini is from a different model family than all evaluation targets, preventing self-preference bias. Its thinking mode supports structured reasoning about whether a response implements a given strategy. The judge receives the

strategy definitions, the declared strategies, and the response text, and produces a score with explicit justification for each strategy.

The judge runs at $T = 1.0$ rather than the conventional $T = 0.0$ because Gemini 3 models are architecturally optimised for their default temperature. Google's developer documentation warns that lowering the temperature below 1.0 can cause looping or degraded reasoning performance. Empirical evidence supports this choice: Renze (2024) finds no statistically significant impact of temperature changes between 0.0 and 1.0 on LLM problem-solving accuracy. Using Gemini 3.1 Pro as an LLM judge at its default temperature is consistent with recent practice: Liu and Tsai (2026)[1] employ it as an automated evaluator within a multi-agent assessment framework. Evaluation consistency is instead ensured through the structured ternary rubric and explicit scoring format.

Additional bias mitigations follow established recommendations (Zheng et al., 2023): cross-model judging ensures the judge never evaluates its own family, CoT justification requires the judge to explain its rating before assigning a score, and a transparent rubric anchors scores to specific behavioural criteria.

NLI cross-encoders were considered and rejected as an alternative. Testing showed an F1 ceiling of approximately 0.55 on the strategy-response alignment task, insufficient for reliable scoring. Cross-encoders also produce no reasoning trace, making it impossible to audit individual judgements.

Method 3 is labelled exploratory. Judge reliability is itself an open question: the judge's ternary scores are not a ground truth but a computational proxy for clinical fidelity assessment. Validation proceeds through human annotation of a random subset of approximately 50 trials, scoring the same rubric. Cohen's $\kappa$ between human and judge provides a reliability estimate. If $\kappa$ is low, Method 3 results carry the caveat explicitly.

Smoke testing showed alignment scores near ceiling (1.0) across all trials. If the full experiment confirms this pattern, alignment becomes a validation check (models consistently implement what they declare) rather than a comparative metric. This outcome would validate Method 1: if plans are faithfully executed, then Jaccard over plan sets measures genuine decision stability rather than label noise.

[1] Preprint, not yet peer-reviewed.

## 3.8. Model Selection

EU data sovereignty serves as the primary selection criterion for the study's main subject. The deployment context (reDreamAI) requires a model that can be self-hosted within EU data residency boundaries (Dale, 2025). All evaluation targets run in non-thinking mode for fair comparison.

The **primary subject** is Mistral Large 3 (675B Mixture of Experts (MoE), 41B active parameters, Apache 2.0). It is the EU sovereign flagship and the model that reDreamAI would deploy in production. The research question centres on whether this model achieves stability comparable to proprietary alternatives.

Two additional EU-sovereign models complete the Mistral comparison. Mistral Small 4 (119B MoE, 6.5B active parameters, Apache 2.0) is Mistral's newest release and the most efficient MoE in the set, activating fewer parameters per token than any other model. Mistral Small 3.2 (24B dense, Apache 2.0) serves as a predecessor baseline, showing whether the architectural shift from dense to MoE affects stability.

The **proprietary ceiling** comprises two models. GPT-5.4 (non-thinking variant) is the strongest proprietary model without thinking overhead. OpenAI reintroduced user-configurable temperature with the non-reasoning variants of GPT-5.2, making controlled temperature comparison possible. Claude Sonnet 4.6 adds a second proprietary reference point; Anthropic's character training emphasises thoughtful, safety-aware interaction, which may influence therapeutic stability independently of raw capability.

**Open-weight comparators** span three size classes and two architectures:

The selection provides size-class diversity (24B–675B), architectural diversity (dense vs. MoE), and provider diversity across five vendors. Qwen 3.5 and Mistral Small 4 are hybrid-thinking models that support optional reasoning. To maintain fair comparison, reasoning is disabled for all evaluation targets.

DeepSeek V3.2 was initially included as a large MoE comparator but was replaced by Qwen 3.5 397B. DeepSeek applies internal temperature scaling (API $T = 1.0$ maps to internal $T \approx 0.3$), which would have complicated cross-model temperature comparisons. Qwen 3.5 397B occupies the same architectural niche without this complexity. Other models considered and dropped include Mistral Medium 3.1 (closed weights), GLM-5 (a third large MoE adds little signal), Nemotron 3 Super (requires $T = 1.0$ universally, complicating temperature comparison), and GPT-oss (mandatory reasoning mode cannot be disabled).

Table 3.2.: Evaluation targets. All models run in non-thinking mode via OpenRouter.

| Group | Model | Size | Selection rationale |
|---|---|---|---|
| *EU-sovereign (Mistral, Apache 2.0)* | | | |
| Primary | Mistral Large 3 | 675B MoE (41B active) | EU-sovereign flagship. Production deployment candidate |
| EU | Mistral Small 4 | 119B MoE (6.5B active) | Newest Mistral. Fewest active parameters of any MoE in set |
| EU | Mistral Small 3.2 | 24B dense | Dense predecessor to Small 4 |
| *Open-weight comparators* | | | |
| MoE | Qwen 3.5 122B | 122B MoE (10B active) | Peer MoE comparator to Mistral Small 4 |
| MoE | Qwen 3.5 397B | 397B MoE (17B active) | Large MoE comparator to Mistral Large 3 |
| Dense | Qwen 3.5 27B | 27B dense | Same size class as Mistral Small 3.2 |
| Dense | OLMo 3.1 32B | 32B dense | Fully open (weights, data, code). Best provenance |
| Dense | Llama 3.3 70B | 70B dense | Continuity with prior efficacy study |
| *Proprietary ceiling* | | | |
| Ceiling | Claude Sonnet 4.6 | Proprietary | Character-trained for safety-aware interaction |
| Ceiling | GPT-5.4 | Proprietary | Strongest non-thinking proprietary model |

## 3.9. Experimental Conditions

The experiment follows a full factorial design:

Rather than testing only two temperature conditions, the experiment uses a five-point temperature sweep that spans from deterministic decoding to the vendor-recommended defaults. The scale is anchored at three points motivated by vendor documentation:

- $T = 0.0$ (greedy decoding): deterministic upper bound on stability. No reproducibility seed is set, as not all vendors support one. Even where a seed is available, GPU parallelisation and dynamic batching introduce floating-point non-determinism that makes output-level reproducibility best effort at most (Yuan et al., 2025). The twenty-trial repetition design captures this variance directly, following Blair-Stanek and Van Durme (2025), who use the same 20-trial protocol at $T = 0.0$ to quantify LLM stability.
- $T = 0.15$: default temperature for Mistral models, taken from the HuggingFace generation configuration for Mistral Small 3.2.
- $T = 0.6$: default temperature for Qwen 3.5 and OLMo 3.1, also close to the standard Llama chat range (0.6–0.7) and the proprietary defaults (GPT-5.4 and Sonnet 4.6 default to $T = 0.7$).

The intermediate points $T = 0.075$ and $T = 0.3$ fill the gap between Mistral's low default and the Qwen/Llama range. $T = 0.075$ was added after an initial four-point run revealed that several models peak in consistency between $T = 0.0$ and $T = 0.15$, suggesting that a small amount of sampling variance can be beneficial (discussed in section 5.2).

This five-point sweep replaces a conventional binary comparison ($T = 0.0$ vs. a single stochastic point). It reveals the full degradation curve rather than a single gap, and allows each model to be evaluated at and around its vendor-recommended operating temperature.

Table 3.3.: Experimental conditions. Total: $6 \times 5 \times 20 = 600$ trials per model, 6,000 across all ten models.

| Dimension | Values |
|---|---|
| Vignettes | anxious, avoidant, cooperative, resistant, skeptic, trauma |
| Trials | 20 per condition |
| Temperatures | $T \in \{0.0,\ 0.075,\ 0.15,\ 0.3,\ 0.6\}$ |
| Models | 10 evaluation targets (Table 3.2) |

## 3.10. Statistical Analysis

The analysis is primarily descriptive. For each model and temperature condition, central tendency and dispersion are reported for all three evaluation metrics, aggregated across vignettes. Jaccard scores are ordinal (discrete values from a small set), so they are summarised with medians and interquartile ranges. BERTScore F1 and alignment scores are reported with means and standard deviations.

Spearman rank correlations serve two purposes. First, per-model correlations between Jaccard and temperature quantify the temperature sensitivity of each model across the five-point sweep. Second, cross-method correlations (Method 1 vs. Method 2, Method 1 vs. Method 3, Method 2 vs. Method 3) assess whether the three evaluation layers capture redundant or independent information. If they diverge, the clinical implications differ: high plan consistency with low response similarity indicates stable strategy selections but variable execution, while the reverse indicates different strategies producing similar surface outputs. Spearman rather than Pearson is used because Jaccard scores are ordinal and BERTScore distributions may be non-normal.

Kruskal-Wallis tests assess whether models and vignettes differ significantly on each metric. These are exploratory rather than confirmatory: the study is designed to describe stability patterns, not to test a preregistered hypothesis.

No fixed threshold defines "stable enough." Clinical significance thresholds for LLM stability do not yet exist in the literature. Results are interpreted comparatively: model against model, across the temperature sweep. The random Jaccard baseline (expected value for random 2-of-6 selections, approximately 0.2) provides a lower reference point.

# 4. Implementation

This chapter describes how the methods from chapter 3 are realised in software. The pipeline is implemented in Python and structured around three concerns: a provider-agnostic LLM interface, a config-driven experiment runner, and a post-hoc aggregation pipeline. All source code is available in the project repository.

## 4.1. Software Architecture

The pipeline follows an async-first design. Every LLM call is non-blocking, and trials within a condition run concurrently via `asyncio.gather()`. This is a direct consequence of the factorial design: 600 trials per model would be prohibitive if executed sequentially. Parallel execution reduces wall-clock time from hours to minutes per model.

**Configuration.** Two YAML files define the full experiment declaratively. `models.yaml` specifies provider endpoints, model identifiers, role assignments, and evaluation targets. `experiment.yaml` controls sampling parameters (trial count, temperatures, maximum token length), metric selection, and output paths. Changing the evaluation target or temperature requires editing a single YAML field rather than modifying code. This separation ensures that experiment conditions are reproducible: the configuration snapshot saved with each run fully specifies how to recreate it.

**Data validation.** Pydantic models enforce type safety at every data boundary. The `Message` and `Conversation` classes validate all fields on construction and serialisation. When a frozen history is loaded from disk, `Conversation.from_dict()` uses `model_validate()` with `extra="ignore"`, silently discarding metadata fields that do not belong to the conversation schema. This is intentional: frozen histories may carry generation-time metadata (timestamps, model versions) that is irrelevant to evaluation.

The same validation applies to plan extraction. The `extract_plan_strategies()` function parses the `<plan>` block from each trial output and validates each strategy

Table 4.1.: Package structure. Each package maps to a distinct concern in the pipeline.

| Package | Responsibility |
|---|---|
| `src/agents/` | Agent implementations (patient, therapist, router). Each agent extends a `BaseAgent` class that handles message formatting, system prompt injection, and provider delegation |
| `src/core/` | Data models (`Message`, `Conversation`), stage and language enums, YAML and prompt loaders |
| `src/llm/` | Provider abstraction. Single-class interface to all LLM backends |
| `src/stacks/` | Orchestration. `GenerationStack` runs the three-agent dialogue loop. `EvaluationStack` runs fused plan-and-response generation from frozen histories |
| `src/evaluation/` | Experiment runner, multi-trial sampler, and all three evaluation metrics (Jaccard, BERTScore, LLM judge alignment) |

identifier against the taxonomy. A malformed plan that cannot be parsed into valid strategy identifiers fails at extraction time rather than propagating downstream where it would silently corrupt Jaccard scores.

**Provider abstraction.** A single `LLMProvider` class wraps an `AsyncOpenAI` client and exposes one async method: `generate(messages, **kwargs)`, returning a tuple of response content and token usage. All ten evaluation targets, the patient agent, and the router use OpenRouter through this interface. The Gemini judge uses Google AI Studio through the same interface, since Google AI Studio exposes an OpenAI-compatible endpoint. Adding a new model or provider requires a configuration entry, not a code change.

A factory function (`create_provider()`) resolves role names (e. g. "therapist", "judge") to concrete provider configurations by looking up the active model assignment in `models.yaml`. For experiment-tier runs, the judge automatically upgrades from Gemini Flash Lite (used during development) to Gemini 3.1 Pro.

The codebase is organised into five packages:

## 4.2. Experiment Execution

An `ExperimentRun` object manages a single evaluation run: one model, one vignette, one temperature, twenty trials. On initialisation it creates a timestamped output directory and saves the configuration snapshot and frozen history.

Execution proceeds in three phases:

1. **Sampling.** A `Sampler` wraps the `EvaluationStack` and runs twenty independent trials. Each trial calls the therapist model once (fused mode), producing a `<plan>` block and a therapeutic response. Trials run in parallel via `asyncio.gather()` with built-in retry logic for rate-limit errors (HTTP 429) using exponential backoff.

2. **Metric computation.** After all trials complete, the run computes all three evaluation metrics. Method 1 (Jaccard) and Method 2 (BERTScore) are computed locally. Method 3 (alignment) issues parallel async calls to the Gemini judge, with a 120-second timeout per trial to handle unresponsive judge calls gracefully.

3. **Persistence.** Results are saved to the run directory.

Each run produces a self-contained output bundle:

```
experiments/runs/{timestamp}_{model}_{vignette}/
    config.yaml              # run parameters (model, temp, slice)
    frozen_history.json      # input conversation (irreplaceable)
    trials/
        trial_01.json        # plan + response + token usage
        ...
        trial_20.json
    metrics.json             # Jaccard, BERTScore, validity, alignment
    judgments.json            # raw judge outputs with reasoning
```

The frozen history is the only irreplaceable artefact in this structure. If the evaluation logic changes (a bug fix in Jaccard computation, a new judge prompt), metrics and judgements can be recomputed from the saved trials. Trials themselves can be recomputed from the frozen history plus the configuration, though at the cost of additional Application Programming Interface (API) calls.

The plan extraction logic handles several output formats that different models produce. The primary pattern matches a closed `<plan>` tag. Fallback patterns handle unclosed tags, bare tag-style strategy names, and plain-text first-line declarations. This robustness is necessary because the ten evaluation targets do not all follow the structured output format identically, despite receiving the same system prompt.

## 4.3. Aggregation Pipeline

After all experiment runs complete, an aggregation script collects results into a single analysis frame. The script iterates over all run directories, loads each `config.yaml` and `metrics.json`, and produces one row per run indexed by model, vignette, slice, and temperature.

Strategy counts are expanded into six separate columns (one per taxonomy category), enabling per-strategy frequency analysis. Alignment scores per strategy are similarly expanded. The resulting DataFrame is saved as a CSV file (`stats/data/all_runs.csv`) that serves as the input for all downstream analysis and visualisation scripts.

This unified frame is a prerequisite for the cross-method analysis in section 5.5. Spearman correlations between Method 1, Method 2, and Method 3 require all three metrics aligned on the same condition index. Without a single frame that maps each condition to its Jaccard, BERTScore, and alignment scores, cross-method comparison would require manual alignment across separate output files.

Downstream scripts in `stats/scripts/` consume this CSV to produce descriptive statistics (`descriptives.py`), statistical tests (`tests.py`), and all five result figures. The separation between aggregation and analysis allows each step to be rerun independently: if a figure script is modified, only visualisation reruns, not the full aggregation.

# 5. Results

All results are reported at the second rewriting-turn boundary (`slice_2`), aggregated across six vignettes. Each data point represents the mean of twenty independent trials. Full per-condition tables appear in Appendix D.

## 5.1. Plan Validity and Strategy Distribution

Before analysing stability, the plan extraction step must be validated. A trial is valid if the model produces a well-formed plan block containing at least one recognised strategy category. Across all 6,000 trials, validity rates exceed 99% for every model. The lowest rates belong to Qwen 3.5 27B (99.3%) and OLMo 3.1 32B (99.7%). All other models achieve 99.8% or above. Validity does not vary meaningfully with temperature. These rates confirm that the fused chain-of-thought prompt reliably elicits structured plan output, and that the subsequent stability metrics are computed over near-complete data.

Strategy distributions vary across models (Figure 5.1). All ten models draw from the same six-category taxonomy, but the relative frequencies differ. `Confrontation` and `self_empowerment` dominate across most models. GPT-5.4 and Sonnet 4.6 show the narrowest distributions, concentrating on two to three categories. Qwen 3.5 27B and Llama 3.3 70B spread selections more evenly across categories. These distributional differences are a prerequisite for interpreting the consistency results that follow: a model that always selects the same two strategies will score high on Jaccard regardless of temperature, while a model that draws from a wider repertoire has more room to vary.

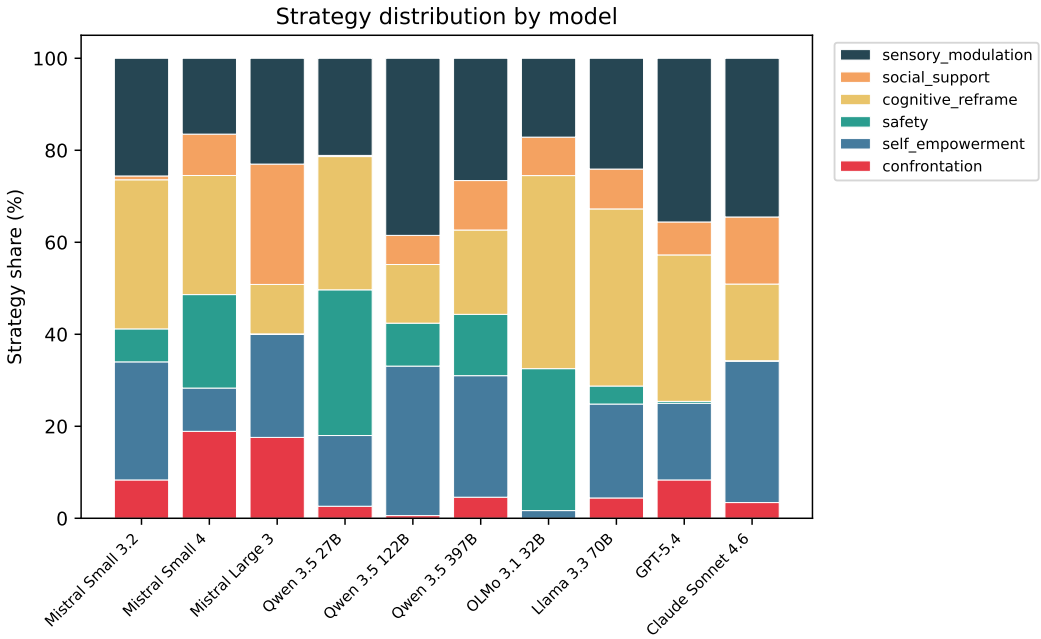


Figure 5.1.: Strategy distribution by model, pooled across temperatures and vignettes. Each bar shows the share of trials selecting each taxonomy category. Models with narrower distributions have less room to vary.

## 5.2. Plan Consistency (Method 1)

Figure 5.2 presents the headline result: mean Jaccard similarity by model and temperature, grouped by model family.

**Proprietary models are temperature-immune.** GPT-5.4 and Sonnet 4.6 achieve $J = 1.000$ at every temperature point. Sampling temperature does not affect their strategy selections at all.

**Most open-weight models peak at low temperatures.** The Mistral family shows a characteristic pattern: Mistral Small 3.2 maintains $J = 1.000$ from $T = 0.0$ through $T = 0.3$ and drops only at $T = 0.6$ ($J = 0.933$). Mistral Large 3 scores $J = 0.933$ at $T = 0.0$, rises to $J = 1.000$ at $T = 0.075$ and $T = 0.15$, then declines to $J = 0.652$ at $T = 0.6$. Mistral Small 4 shows the most pronounced version of this pattern: $J = 0.722$ at $T = 0.0$, jumping to $J = 1.000$ at $T = 0.075$ before declining.

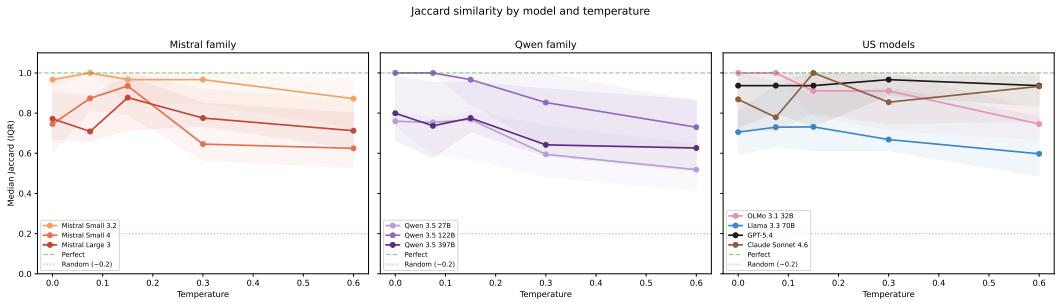


Figure 5.2.: Mean Jaccard similarity by model and temperature, grouped by family. Shaded bands show ±1 SD across vignettes. The dashed line marks the random baseline ($J \approx 0.2$). Higher is more consistent.

**Greedy decoding is not always optimal.** Three models score higher at $T = 0.075$ than at $T = 0.0$: Mistral Small 4 ($0.722 \rightarrow 1.000$), Mistral Large 3 ($0.933 \rightarrow 1.000$), and Llama 3.3 70B ($0.756 \rightarrow 0.881$). A small amount of sampling variance produces more consistent strategy selection than pure greedy decoding for these models.

**Temperature sensitivity varies by model.** Per-model Spearman correlations between Jaccard and temperature (Table 5.1) reveal three groups. OLMo 3.1 32B ($\rho = -0.632$, $p < 0.001$), Qwen 3.5 27B ($\rho = -0.576$, $p = 0.001$), and Qwen 3.5 122B ($\rho = -0.513$, $p = 0.009$) show statistically significant degradation. All other models, including the entire Mistral family, show no significant temperature effect on Jaccard. The pooled correlation across all models is $\rho = -0.228$ ($p < 0.001$).

Table 5.1.: Spearman correlation between Jaccard and temperature, per model. Significant correlations ($p < 0.05$) are marked with an asterisk.

| Model | $\rho$ | $p$ | Sig. |
|---|---|---|---|
| Mistral Small 3.2 | −0.190 | 0.315 | |
| Mistral Small 4 | −0.196 | 0.299 | |
| Mistral Large 3 | −0.277 | 0.139 | |
| Qwen 3.5 27B | −0.576 | 0.001 | * |
| Qwen 3.5 122B | −0.513 | 0.009 | * |
| Qwen 3.5 397B | −0.232 | 0.217 | |
| OLMo 3.1 32B | −0.632 | <0.001 | * |
| Llama 3.3 70B | −0.153 | 0.421 | |
| GPT-5.4 | +0.075 | 0.695 | |
| Claude Sonnet 4.6 | −0.063 | 0.741 | |
| Pooled | −0.228 | <0.001 | * |

## 5.3. Response Similarity (Method 2)

BERTScore F1 measures semantic similarity between response texts, independent of the plan block. The pooled temperature effect is stronger than for Jaccard ($\rho = -0.486$, $p < 0.001$): rising temperature changes response wording more than it changes strategy selection.

However, BERTScore has limited sensitivity to plan-level differences. The slice depth analysis (section 5.6) confirms this: across conversation depths, Jaccard varies visibly while BERTScore remains flat. A model that switches from `confrontation` to `cognitive_reframe` changes its therapeutic plan but produces surface text similar enough that BERTScore does not detect the difference.

BERTScore therefore captures response *style* consistency rather than decision *content* consistency. It serves as a sanity check: if BERTScore were low, the model would be generating erratic text regardless of strategy. The substantive signal for therapeutic stability comes from Jaccard and alignment.

Full BERTScore results by model and temperature appear in Appendix D.

## 5.4. Plan-Response Alignment (Method 3)

Figure 5.3 shows plan-response alignment by temperature and per model. Alignment measures whether the model's response implements the strategies it declared in its plan.

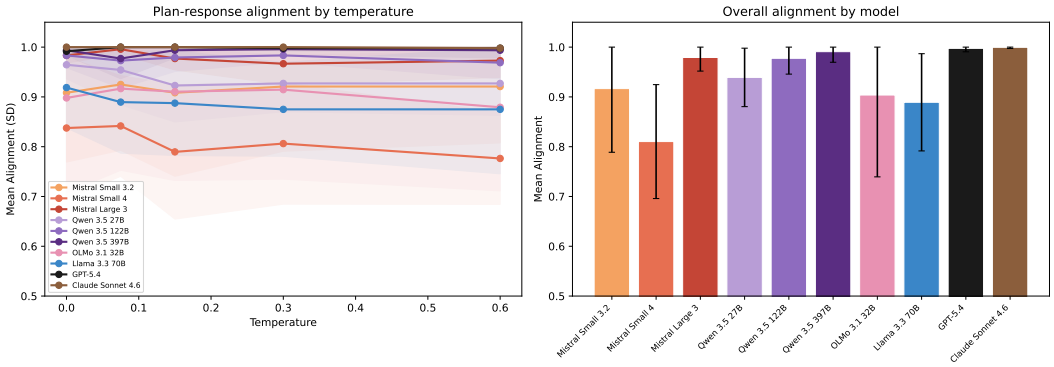


Figure 5.3.: Plan-response alignment by temperature (left) and overall by model (right). Error bars show ±1 SD. Most models implement their declared strategies consistently regardless of temperature.

The central finding is that alignment is temperature-independent. The pooled Spearman correlation between alignment and temperature is $\rho = -0.053$ ($p = 0.361$), not significant. Models do not get worse at implementing their declared strategies as temperature rises. They select *different* strategies more often (lower Jaccard) but implement whichever strategies they select equally well (stable alignment).

This dissociation has a clinical implication: temperature-induced instability occurs at the decision-making level, not at the execution level. A model at high temperature does not produce incoherent responses. It produces coherent responses to a different therapeutic plan.

Kruskal-Wallis tests confirm that models differ significantly on alignment ($p < 0.001$). Mistral Small 4 shows the lowest mean alignment, consistent with its erratic strategy selection at extreme temperatures. The proprietary models and Mistral Small 3.2 cluster near the top.

## 5.5. Cross-Method Analysis

The three evaluation methods are related but not redundant. Table 5.2 reports Spearman correlations between all method pairs, and Figure 5.4 visualises the relationship between strategy and semantic consistency.

The moderate Jaccard–BERTScore correlation ($\rho = 0.560$) indicates that models with more consistent strategy selection also tend to produce more similar response text, but the relationship is not strong enough for one metric to substitute for the

Table 5.2.: Spearman correlations between the three evaluation methods. All correlations are significant ($p < 0.001$). The moderate Jaccard–BERTScore correlation and weak Jaccard–Alignment correlation confirm that the methods capture distinct dimensions.

| Method pair | $\rho$ | Interpretation |
|---|---|---|
| Jaccard vs. BERTScore | 0.560 | Moderate |
| Jaccard vs. Alignment | 0.285 | Weak |
| BERTScore vs. Alignment | 0.376 | Weak–moderate |

other. The weak Jaccard–Alignment correlation ($\rho = 0.285$) confirms that plan consistency and plan-response coherence measure different properties.

The scatter plot (Figure 5.4) reveals the model landscape. GPT-5.4 and Sonnet 4.6 anchor the upper-right corner: high on both dimensions. Qwen 3.5 27B sits in the lower-left: low strategy consistency and low semantic similarity. Mistral Small 4 presents an interesting case: moderate Jaccard but the lowest BERTScore in the set, suggesting that even when it selects the same strategies, its response wording varies considerably.

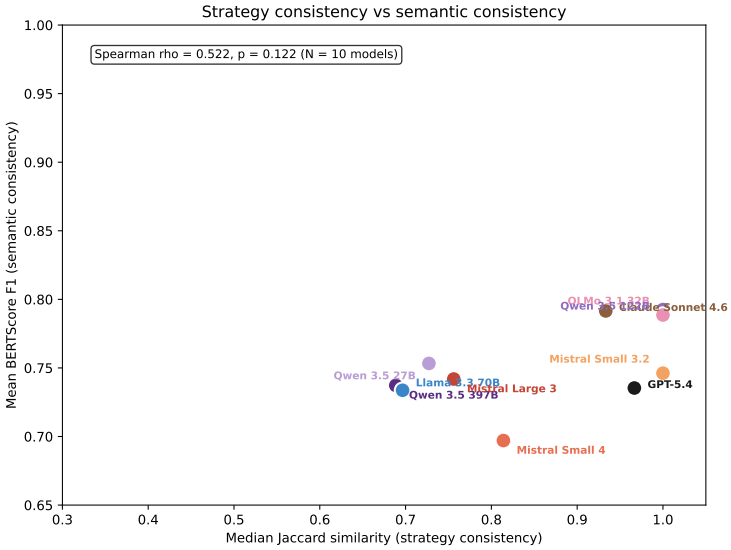


Figure 5.4.: Strategy consistency (median Jaccard) versus semantic consistency (mean BERTScore F1) per model, pooled across temperatures. The proprietary models cluster in the upper-right corner. Qwen 3.5 27B and Mistral Small 4 occupy opposite ends of the BERTScore axis despite similar Jaccard ranges.

## 5.6. Secondary Effects

### 5.6.1. Vignette Effects

Kruskal-Wallis tests detect a significant vignette effect on Jaccard ($p = 0.007$) and BERTScore ($p = 0.015$). Some patient profiles produce more consistent therapeutic responses than others across models. Full per-vignette heatmaps appear in Appendix D.

### 5.6.2. Conversation Depth

A preliminary analysis tested Mistral Large 3 at five conversation depths (slices 1–5) across all six vignettes at $T = 0.075$ and $T = 0.15$. Figure 5.5 shows the results.

Jaccard consistency is lowest at the second slice (mean $J = 0.77$ at $T = 0.075$) and rises to near-ceiling values from the third slice onward ($J > 0.93$). BERTScore and alignment remain flat across all depths, confirming that conversation depth affects strategy *selection* but not response *wording* or plan *execution*.

The main experiment evaluates at slice 2, making the reported stability scores a conservative estimate. A model that achieves high consistency at this depth would

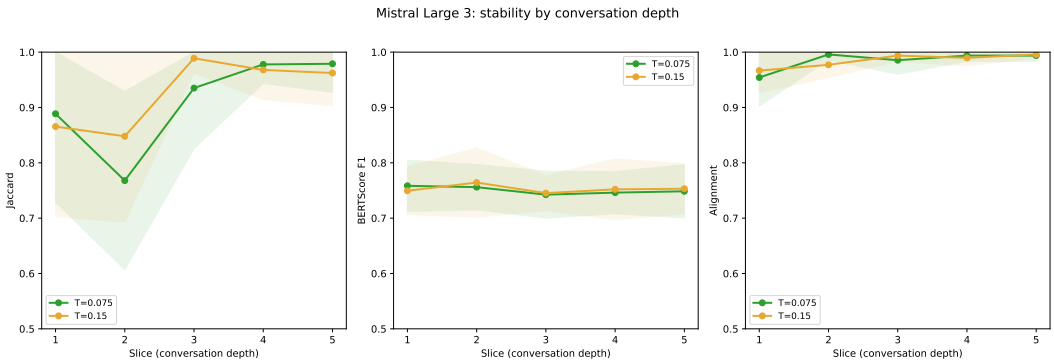


Figure 5.5.: Stability by conversation depth for Mistral Large 3. Jaccard is lowest at slice 2 and rises to near-ceiling values from slice 3 onward. BERTScore and alignment remain flat across depths.

likely score even higher at later conversation points where prior exchanges provide more anchoring context.

# 6. Discussion

## 6.1. Interpreting the Three-Level Framework

The three evaluation methods were designed to capture distinct aspects of therapeutic stability (section 3.7). The results confirm that they do: the cross-method correlations are moderate at best ($\rho = 0.560$ for Jaccard–BERTScore, $\rho = 0.285$ for Jaccard–Alignment). Had any two methods been strongly correlated, one would be redundant. Instead, each method reveals information that the others miss.

The most informative finding is the dissociation between plan consistency and alignment. Temperature increases reduce Jaccard (models select different strategies) but do not reduce alignment (models implement whichever strategy they select equally well). This pattern localises the source of instability to the *decision-making* layer rather than the *execution* layer. A model at high temperature does not produce incoherent or unfaithful responses. It produces coherent responses to a different therapeutic plan.

This distinction has practical value as a diagnostic tool. If a deployment shows unstable therapeutic behaviour, the framework can identify where the instability originates. High Jaccard with low alignment would indicate that the model selects consistent strategies but fails to implement them (an execution problem). Low Jaccard with high alignment, which is the dominant pattern observed here, indicates that the model implements its plans faithfully but selects different plans across runs (a decision problem). Each pattern implies a different remediation strategy: execution problems suggest prompt engineering or fine-tuning, while decision problems suggest temperature tuning or strategy constraints.

BERTScore occupies a more limited role than initially anticipated. It correlates moderately with Jaccard but cannot distinguish between clinically different plans when the surface text remains similar. The slice depth analysis confirms this: Jaccard varies across conversation depths while BERTScore stays flat (Figure 5.5). A model that switches from `confrontation` to `cognitive_reframe` changes its therapeutic approach, but the response text remains similar enough that BERTScore

does not detect the difference. BERTScore therefore serves primarily as a sanity check for response coherence rather than as an independent stability measure.

## 6.2. Cross-Model Comparison

The central question of this thesis is whether the EU-sovereign flagship, Mistral Large 3, achieves therapeutic stability comparable to proprietary alternatives (Figure 5.2).

At the vendor-recommended temperature ($T = 0.15$), Mistral Large 3 achieves $J = 1.000$, matching GPT-5.4 and Sonnet 4.6. At its peak ($T = 0.075$), it also reaches $J = 1.000$. The sovereignty tradeoff, at low temperatures, is negligible: the EU flagship matches the proprietary ceiling on plan consistency.

The tradeoff becomes visible at higher temperatures. At $T = 0.6$, Mistral Large 3 drops to $J = 0.652$, while the proprietary models maintain $J = 1.000$. However, $T = 0.6$ is well above Mistral's recommended operating range. A deployment that follows vendor guidance ($T \leq 0.15$) would not encounter this degradation.

Mistral Small 3.2 (24B dense) is the most stable open-weight model in the set, maintaining $J = 1.000$ through $T = 0.3$ and showing no statistically significant temperature sensitivity.

Mistral Small 4 presents the most pronounced version of the greedy-suboptimality pattern. At $T = 0.0$ it scores only $J = 0.722$, then jumps to $J = 1.000$ at $T = 0.075$. A small amount of sampling variance helps the model settle on a consistent strategy rather than oscillating between candidates under greedy decoding. Mistral Large 3 ($0.933 \rightarrow 1.000$) and Llama 3.3 70B ($0.756 \rightarrow 0.881$) show the same direction.

Among the open-weight comparators, parameter count does not straightforwardly predict stability. Qwen 3.5 122B (10B active) outperforms Qwen 3.5 397B (17B active) at every temperature point. OLMo 3.1 32B matches larger models at low temperatures but degrades significantly at higher ones ($\rho = -0.632$, strongest temperature sensitivity in the set). The Qwen 3.5 family shows a consistent pattern across its three size classes: the mid-size 122B model is the most stable, while the smallest (27B) and largest (397B) are less consistent.

The proprietary models (GPT-5.4, Sonnet 4.6) form a distinct tier. Their perfect Jaccard across all temperatures suggests that character training, safety-layer alignment, or other proprietary techniques create a stability floor that sampling temperature cannot disrupt. Whether this reflects genuine robustness or a narrower

output distribution (fewer strategies available to vary) cannot be determined from these results alone.

## 6.3. Temperature and Clinical Deployment

The five-point temperature sweep reveals that the relationship between temperature and stability is not monotonic for all models. Three models achieve strictly higher Jaccard at $T = 0.075$ than at $T = 0.0$, meaning greedy decoding is actively counterproductive for them.

Three models achieve their peak Jaccard at $T = 0.075$ rather than $T = 0.0$: Mistral Small 4 ($0.722 \rightarrow 1.000$), Mistral Large 3 ($0.933 \rightarrow 1.000$), and Llama 3.3 70B ($0.756 \rightarrow 0.881$). This finding has a practical implication: clinicians or engineers deploying these models should not default to $T = 0.0$ for maximum stability. A small amount of sampling variance appears to help the model resolve between competing strategy candidates, producing more consistent selections than pure greedy decoding.

The pooled Spearman correlation ($\rho = -0.228$, $p < 0.001$, Table 5.1) confirms that higher temperature generally reduces plan consistency, but the effect is weak. Per-model analysis reveals that only three models show statistically significant degradation: OLMo 3.1 32B, Qwen 3.5 27B, and Qwen 3.5 122B. The remaining seven models, including the entire Mistral family, show no significant temperature effect on Jaccard.

For clinical deployment, these results suggest a practical recommendation: operate at or near the vendor-recommended temperature. Models designed for a specific temperature range tend to be most stable within it. Pushing a model to $T = 0.0$ in the hope of maximising determinism can be counterproductive.

Alignment is temperature-independent across all models ($\rho = -0.053$, $p = 0.361$). This means that the clinical cost of higher temperature is limited to strategy *selection* variability. The model does not become worse at *executing* its chosen strategy. A patient receiving care from a high-temperature model would encounter different therapeutic approaches across sessions, but each individual session would be internally coherent.

## 6.4. Vignette-Dependent Behaviour

The significant vignette effect on Jaccard ($p = 0.007$) and BERTScore ($p = 0.015$) indicates that some patient profiles are harder to treat consistently than others. This is clinically relevant: if a model is stable for cooperative patients but variable for resistant or trauma-presenting patients, deployment safety depends on the patient population being served.

This finding suggests that stability evaluation should not rely solely on aggregate metrics. Per-vignette reporting, as provided in this study's appendix, is necessary to identify the conditions under which a model's consistency breaks down. A model that appears stable on average may be unreliable for the specific patient profiles that need the most careful treatment.

Full per-vignette analysis is provided in Appendix D.

## 6.5. Evaluation Framework Reflections

**Strengths.** The framework's hierarchical structure proved useful in practice. By measuring stability at three layers, the analysis could identify that instability is concentrated at the decision-making level (variable strategy selection) rather than the execution level (unfaithful implementation). A single-metric evaluation using only BERTScore, for instance, would have missed the plan-level instability entirely, as BERTScore was insensitive to strategy changes (section 5.3).

The frozen history design ensures strict reproducibility: every trial receives identical input, and any observed variance is attributable to the model under test. The 20-trial repetition provides statistical coverage of the output distribution rather than relying on a single evaluation (Blair-Stanek & Van Durme, 2025).

**Taxonomy sensitivity.** The measured consistency scores are sensitive to taxonomy design. A coarser taxonomy (fewer, broader categories) inflates Jaccard scores because more strategy selections fall into the same category. A finer taxonomy deflates them. The six-category taxonomy used here (Table 3.1) represents a balance between clinical granularity and discriminative power, developed through four iterations that resolved a severe distribution skew (section 3.6). Different taxonomy choices would produce different absolute Jaccard scores. The *relative* ordering of models is more robust to taxonomy changes than the absolute values, because all models are evaluated against the same category set.

**Declared intent, not cognition.** The `<plan>` block is framed as declared intent rather than a window into model cognition (section 3.5). This framing has a limitation: it cannot distinguish between a model that genuinely decides on a strategy and then implements it, and a model that generates a plausible-sounding plan after having already committed to a response direction. The high alignment scores observed across most models are consistent with either interpretation. What the plan *can* claim is that strategy declarations vary across runs, and this variation predicts variable therapeutic behaviour regardless of the underlying mechanism.

**Generalisability.** The three-level framework is not specific to IRT. Any manualized therapeutic protocol with identifiable strategy categories could be evaluated using the same structure: extract declared strategies (Method 1), compare response texts (Method 2), and verify plan-response alignment (Method 3). Candidate protocols include CBT (with its established technique taxonomy), dialectical behaviour therapy, and motivational interviewing. The framework would require a new strategy taxonomy for each protocol, but the evaluation infrastructure and metrics transfer directly.

**Ethical scope.** This framework measures whether a model produces *consistent* therapeutic responses, not whether those responses are *clinically appropriate*. A model that consistently selects the wrong strategy scores high on plan consistency. Stability is a necessary precondition for clinical deployment, as stated in section 1.2, but it is not sufficient. All evaluations use synthetic patients and no human participants are involved. Clinical validation through human expert evaluation remains necessary before any deployment to real patients.

**Methodological constraints.** No reproducibility seed was set, as not all vendors support one. Even where seeds are available, they provide best-effort determinism at most (Yuan et al., 2025). The trial repetition design captures stochastic variance directly.

The experiment evaluates at a single conversation depth (`slice_2`), chosen as a conservative estimate based on the preliminary depth analysis (section 3.3). Results at other depths may differ. All models were accessed through OpenRouter, so provider-side batching and load balancing may contribute to the non-determinism observed at $T = 0.0$, though this mirrors realistic deployment conditions.

Further limitations regarding scope, vignette coverage, judge validation, and taxonomy constraints are discussed in section 7.2.

# 7. Conclusion

## 7.1. Summary of Findings

This thesis asked how consistent LLM therapeutic responses are across independent runs within a structured IRT protocol. Three evaluation methods were applied to ten models across 6,000 trials spanning five temperature conditions.

The main findings are:

1. **The EU-sovereign flagship matches proprietary models at recommended temperatures.** Mistral Large 3 achieves $J = 1.000$ at $T = 0.075$ and $T = 0.15$, equal to GPT-5.4 and Sonnet 4.6. The sovereignty trade-off in therapeutic stability is negligible when the model operates within its vendor-recommended range.

2. **Instability is a decision-making problem, not an execution problem.** Temperature reduces plan consistency (Jaccard) but does not reduce plan-response alignment. Models select different strategies at higher temperatures but implement whichever strategy they select equally well.

3. **Greedy decoding is not always optimal.** Three models achieve higher consistency at $T = 0.075$ than at $T = 0.0$. A small amount of sampling variance can help a model settle on a consistent strategy.

4. **The three evaluation levels capture distinct information.** Cross-method correlations are moderate at best ($\rho = 0.285$ to $0.560$). BERTScore is insensitive to plan-level changes. No single metric captures the full picture of therapeutic stability.

5. **Patient profile matters.** A significant vignette effect ($p = 0.007$) means that stability evaluations based on aggregate metrics alone may miss per-profile weaknesses.

Regarding the two contributions stated in section 1.4: the three-level framework (Contribution 1) fulfilled its design goal by identifying that instability is concentrated

at the strategy selection layer, a finding that neither BERTScore nor alignment alone could have revealed. The cross-model comparison (Contribution 2) demonstrates that the EU-sovereign Mistral Large 3 achieves stability comparable to proprietary models at its recommended operating temperature, supporting its viability for clinical deployment under sovereignty constraints.

## 7.2. Limitations

**Scope.** The study evaluates a single phase (rescripting) of a single protocol (IRT). Stability in other IRT stages or other therapeutic protocols may differ. The rescripting phase was selected because it requires active strategy decisions, but this focus limits the generalisability of the absolute scores reported.

**Synthetic patients.** All patient interactions are simulated by an LLM. Synthetic patients follow scripted behavioural profiles and do not exhibit the unpredictable responses of real patients. The vignettes are designed to sample the difficulty range (section 3.2), but six profiles cannot claim exhaustive coverage of clinical variability. A cooperative synthetic patient may underestimate the challenges a model faces with a real patient whose behaviour deviates from any template.

**Ethical considerations.** This study evaluates LLM stability using synthetic patients, not real clinical interactions. No human participants were involved. The results measure computational consistency and do not constitute evidence of clinical safety or efficacy. IRT targets nightmares rather than psychiatric diagnoses, which limits clinical risk relative to full psychotherapy, but any deployment to real patients would require clinical validation beyond what this framework provides.

**Judge validation.** The LLM judge (Gemini 3.1 Pro) has not been validated against human annotations. Method 3 results are a computational proxy for clinical fidelity assessment and should be interpreted accordingly. The planned Cohen's $\kappa$ validation remains outstanding.

**Taxonomy constraints.** The six-category strategy taxonomy constrains what can be measured. If models employ therapeutic strategies that fall outside these categories, that variation is invisible to Method 1. The taxonomy was developed through iterative refinement (section 3.6), but different category boundaries would produce different absolute consistency scores.

**Consistency is not quality.** A model that consistently selects an inappropriate strategy scores high on plan consistency. Stability is a necessary but not sufficient condition for clinical deployment.

## 7.3. Future Work

Several directions follow from this work.

**Full protocol evaluation.** Extending the evaluation beyond the rescripting phase to all five IRT stages would test whether the stability patterns observed here hold when the therapist model faces different types of clinical decisions (listening, summarising, rehearsing).

**Other manualized protocols.** The three-level framework transfers to any protocol with identifiable strategy categories. CBT, dialectical behaviour therapy, and motivational interviewing are natural candidates. Each would require a domain-specific taxonomy but could reuse the evaluation infrastructure.

**Longitudinal drift.** Model vendors update weights and serving infrastructure without notice. Repeating the evaluation across model versions would reveal whether stability improves, degrades, or fluctuates across release cycles.

**Multi-language evaluation.** reDreamAI targets German-speaking users. Evaluating therapeutic stability in German would test whether the patterns observed in English transfer across languages, or whether language-specific training data introduces different stability profiles.

**Clinical validation.** Human therapist ratings on the same ternary rubric would validate Method 3 and provide an external ground truth for the computational framework. Comparing human and LLM judge scores would quantify the reliability of automated fidelity assessment.

# Acknowledgements

# Use of AI Tools in This Thesis

Throughout the research and writing of this thesis, I made use of several LLM-based tools in a supporting capacity.

For intellectual engagement with the subject matter, I used Claude (Anthropic)[1] and ChatGPT (OpenAI)[2] as discussion partners. Their role was to challenge my reasoning, surface counterarguments, and suggest alternative framings for the arguments I had already developed. As English is not my native language, I occasionally used these tools to refine phrasing where my original formulations lacked precision or fluency.

For software development, I used Claude Code and Cursor (with Claude Sonnet) as coding assistants for the evaluation pipeline that forms the core of this thesis. Given that the pipeline itself measures LLM behavioral stability, I took particular care to verify all AI-generated code before inclusion, to avoid introducing confounds into the instrument being studied.

No part of the thesis text was generated by AI. All arguments, interpretations, and conclusions are my own.

[1] https://claude.ai
[2] https://chatgpt.com

# Glossary

**API** Application Programming Interface

**BDI** Brief Dreamer Interview

**CBT** Cognitive Behavioral Therapy

**CoT** Chain-of-Thought

**EU** European Union

**GDPR** General Data Protection Regulation

**IRT** Imagery Rehearsal Therapy

**JSON** JavaScript Object Notation

**LLM** Large Language Model

**MoE** Mixture of Experts

**NLI** Natural Language Inference

**NLP** Natural Language Processing

**PTSD** Post-Traumatic Stress Disorder

**RLHF** Reinforcement Learning with Human Feedback

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# A. Strategy Taxonomy

# B. Prompts

# C. Architecture Diagrams

# D. Supplementary Results